PRACTICE 5A: SOFTWARE TESTING

1. Objectives

- Learn to design unit test cases applying black box and white box techniques for an object-oriented software system.
- Learn to run unit tests using a testing framework, in this case formed by the *JUnit* and *Eclemma tools.*
- Learn how to run integration tests on Swing-based graphical interface applications using the *AssertJ Swing library.*

2. Activities

The student must carry out the (partial) process of designing and executing unit and integration tests for the application that was developed in practices 2 and 3. For black box testing, the equivalent partitioning and AVL techniques must be applied, and for white box testing, decision/ condition and statement coverage is required.

NOTE: For the development of this practice, a copy of the projects developed in Practice 3 must be made in the folder corresponding to Practice 5A and work on them.

The steps to follow to carry out the testing process are as follows:

1. Unit tests of the Employee class.
   i. Design test cases for both classes (excluding get/set methods).
   ii. Write the code for the corresponding test class using JUnit.
   iii. Run the test class and correct defects.
   iv. When the black-box tests are successful, check the coverage. In this case, it is necessary to achieve full coverage, including get/set methods.
   v. Design new test cases to achieve the desired coverage if not to have been achieved with the black box tests.

   NOTE: During the unit testing phase of the application, the classes of the DAO, business, and presentation layers should be tested. However, their development is not covered in this exercise.

2. Integration tests of the presentation layer with the business layer and the DAO layer.
   Design and implement integration tests to verify the joint operation of the VistaGerente class with the rest of the application layers.
   i. Design test cases for the functionality implemented so far in the class (corresponding to the Store Inquiry use case).
   ii. Implement in JUnit the test cases designed using AssertJ.
   iii. Execute the test cases and correct errors.
   iv. When all black box tests are correct, check the coverage.

v. Design new test cases to achieve the desired coverage if not to have been achieved with the black box tests.

Since these are integration tests, to distinguish them from unit tests, the test class name must include the string "IT" and they are executed in the verify phase of Maven, not the test phase. Additionally, the Maven-failsafe-plugin must be explicitly included in the pom.xml file for them to function (see T15 of the Maven Seminar).

Note: For instructions on how to complete this exercise, please refer to Annex 1.

3. Evaluation Criteria and Clarifications

A compressed file containing the following elements must be submitted:

- Complete Maven projects, with all classes involved (both application and as a test), duly exported.
- A report summarizing the applied testing process, identifying the phases, the techniques used in each phase, and the test case design process.

The following will be taken into account:

a) The report format.
b) The correction of the testing process of the domain classes.
c) The correction of the testing process of the VistaAgente class.

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APPENDIX 1. IMPLEMENTATION OF THE PERSISTENCE LAYER

The persistence layer, or DAO layer, is implemented using H2, a Java-based database that can be configured as an in-memory database, which is how it has been used in this application. This means that the data only exists for the lifetime of the application.

The initial state of the database when the application is launched, which is necessary to know in order to carry out the verification and validation phase, is shown in Table 1.

Table 1. Initial state of the database

Stores							
ID	Name	Address	Employees				
			ID card	Name	Low	Date	Category
1	Store A	Address A	11111111A	Juan Pérez	False	2002-01-15	MANAGER
			11111111B	Maria Rodriguez	True	2016-05-20	SALESPERSON
			11111111C	Luis Martínez	False	2022-05-21	AUXILIARY
2	Store B	Address B	11111111D	Lucía Ibáñez	False	2010-06-01	IN CHARGE
3	Store C	Address C					